WEEKLY REPORT 3

CSE623: Machine Learning Theory And Practice

Crop Yield Prediction Using Classical Machine Learning Models and Climatic Factors

Ahmedabad University

The Visionary

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1 Introduction

During this week, the focus shifted from data preprocessing and exploratory data analysis to the implementation of machine learning models for crop yield prediction. After preparing the dataset in the previous phase, the cleaned dataset was used to train multiple regression-based machine learning models.

The goal of this phase was to develop predictive models capable of estimating crop yield using agricultural and climatic factors such as rainfall, fertilizer usage, pesticide usage, and cultivated area. Different classical machine learning algorithms were implemented and trained using the prepared dataset.

2 Work Done This Week

The following tasks were completed during Week 3:

- Preparing the dataset for machine learning model training.
- Splitting the dataset into training and testing subsets.
- Implementing classical regression models including Linear Regression.
- Implementing tree-based machine learning models such as Decision Tree Regressor.
- Implementing ensemble learning models including Random Forest Regressor and Gradient Boosting Regressor.
- Training the implemented models using the training dataset.
- Evaluating model performance using standard regression evaluation metrics.

3 Dataset Splitting

Before training the machine learning models, the dataset was divided into two subsets: training data and testing data. This step ensures that the models are evaluated on unseen data, which helps measure their generalization capability.

An 80:20 split ratio was used:

- **Training Set (80%)** – Used to train the machine learning models.
- **Testing Set (20%)** – Used to evaluate model performance.

The training dataset allows the models to learn patterns from the agricultural features, while the testing dataset helps assess the predictive performance of the trained models.

4 Machine Learning Models Implemented

Several classical machine learning regression models were implemented to predict crop yield.

4.1 Linear Regression

Linear Regression is one of the most fundamental regression techniques used for predictive modeling. It establishes a linear relationship between input features and the target variable. In this project, Linear Regression was used as a baseline model to evaluate how well crop yield can be predicted using agricultural parameters.

4.2 Decision Tree Regressor

Decision Tree Regressor is a tree-based machine learning model that splits the dataset into smaller subsets based on feature values. Each node in the tree represents a decision rule, allowing the model to capture non-linear relationships between features and crop yield.

4.3 Random Forest Regressor

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. The final prediction is obtained by averaging the predictions from multiple trees.

4.4 Gradient Boosting Regressor

Gradient Boosting is another ensemble learning technique where models are trained sequentially. Each new model focuses on correcting the errors made by the previous model, resulting in improved prediction performance.

5 Model Training

After implementing the machine learning algorithms, the models were trained using the training dataset. During this phase, the algorithms learned relationships between the input variables such as rainfall, fertilizer usage, pesticide usage, and cultivated area with the target variable, which is crop yield.

The trained models were then used to generate predictions on the testing dataset to evaluate their performance.

6 Model Evaluation

To measure the performance of the trained models, several regression evaluation metrics were used.

6.1 Mean Absolute Error (MAE)

Mean Absolute Error calculates the average absolute difference between the predicted yield and the actual yield. Lower MAE values indicate better model performance.

6.2 Mean Squared Error (MSE)

Mean Squared Error measures the average squared difference between predicted and actual values. This metric penalizes larger errors more heavily.

6.3 Root Mean Squared Error (RMSE)

Root Mean Squared Error is the square root of the Mean Squared Error and provides an error value in the same unit as the predicted variable.

6.4 R-Squared Score

The R-squared score indicates how well the model explains the variance in the crop yield data. Higher R-squared values indicate better predictive performance.

7 Work to be Done Next Week

The following tasks are planned for the next phase of the project:

- Performing comparative analysis of the implemented machine learning models.
- Identifying the best-performing model based on evaluation metrics.
- Conducting feature importance analysis to determine which factors most strongly influence crop yield.
- Generating additional visualizations to interpret model predictions.
- Preparing the final analysis and result interpretation for the project.